EE 121B: Principles of Semiconductor Device Design

Lecture 1:

Introduction

Semiconductor Fundamentals

Sam Emaminejad

Objectives

- Developing understanding of the principles of operation of important semiconductor devices including PN junction, MOS transistor, and bipolar transistor.
- Developing generalizable analytical skills to design and evaluate the performance of semiconductor devices.

Workload and Grading Scheme

- 6 Homework: 20%
  - Lowest homework score will be dropped
- Midterm 1: 20%
- Midterm 2: 20%
- Final: 40%
- Survey and Class Participation: 1%

* Bonus points for in-class discussions

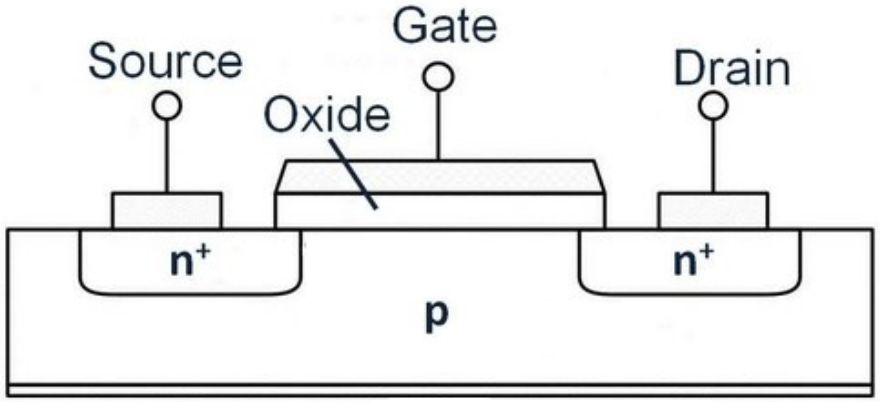
Reading Material

- **References**:
  - *Modern Semiconductor Devices for Integrated Circuits* by C. Hu (Prentice Hall, 2009)
  - *Semiconductor Device Fundamentals* by R. F. Pierret (Addison Wesley, 1996)

Course Overview

1. Semiconductor Properties
2. P-N Junction
3. Metal-Semiconductor Contacts
4. MOS Devices
5. Bipolar Junction Transistor

Metal-Oxide-Semiconductor (MOS)
Field-Effect Transistor (FET)



http://qph.is.quoracdn.net/main-qimg-38a99b4f215e161bf31964795cfce7d2?convert_to_webp=true

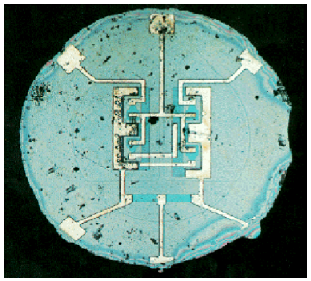
The Integrated Circuit (IC)

- **An IC consists of interconnected electronic components in a single piece ("chip") of semiconductor material.**
  - **In 1958, Jack S. Kilby (*Texas Instruments*) showed that it was possible to fabricate a simple IC in germanium.**



Kilby's first integrated circuit in germanium.

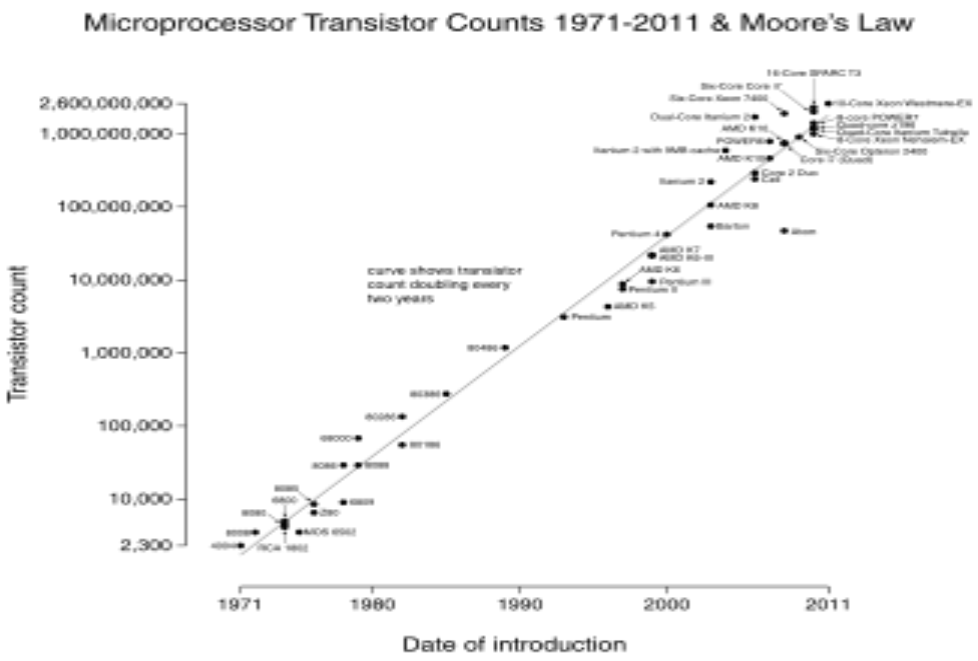
  - **In 1959, Robert Noyce (*Fairchild Semiconductor*) demonstrated an IC made in silicon using SiO_2 as the insulator and Al for the metallic interconnects.**



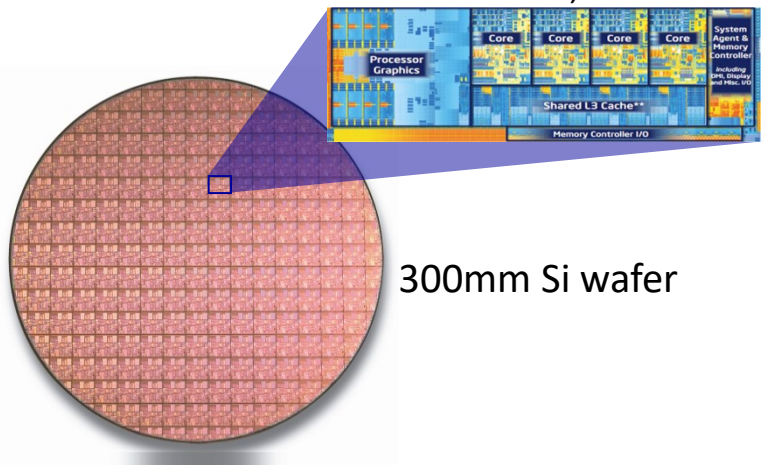
The first planar IC
(actual size: 0.06 in. diameter)

From a Few, to Billions of Components

- **By connecting a large number of components, each performing simple operations, an IC that performs complex tasks can be built.**
- **The degree of integration has increased at an exponential pace over the past ~50 years.**



Intel Ivy Bridge Processor
1.4B transistors, 160 mm^2

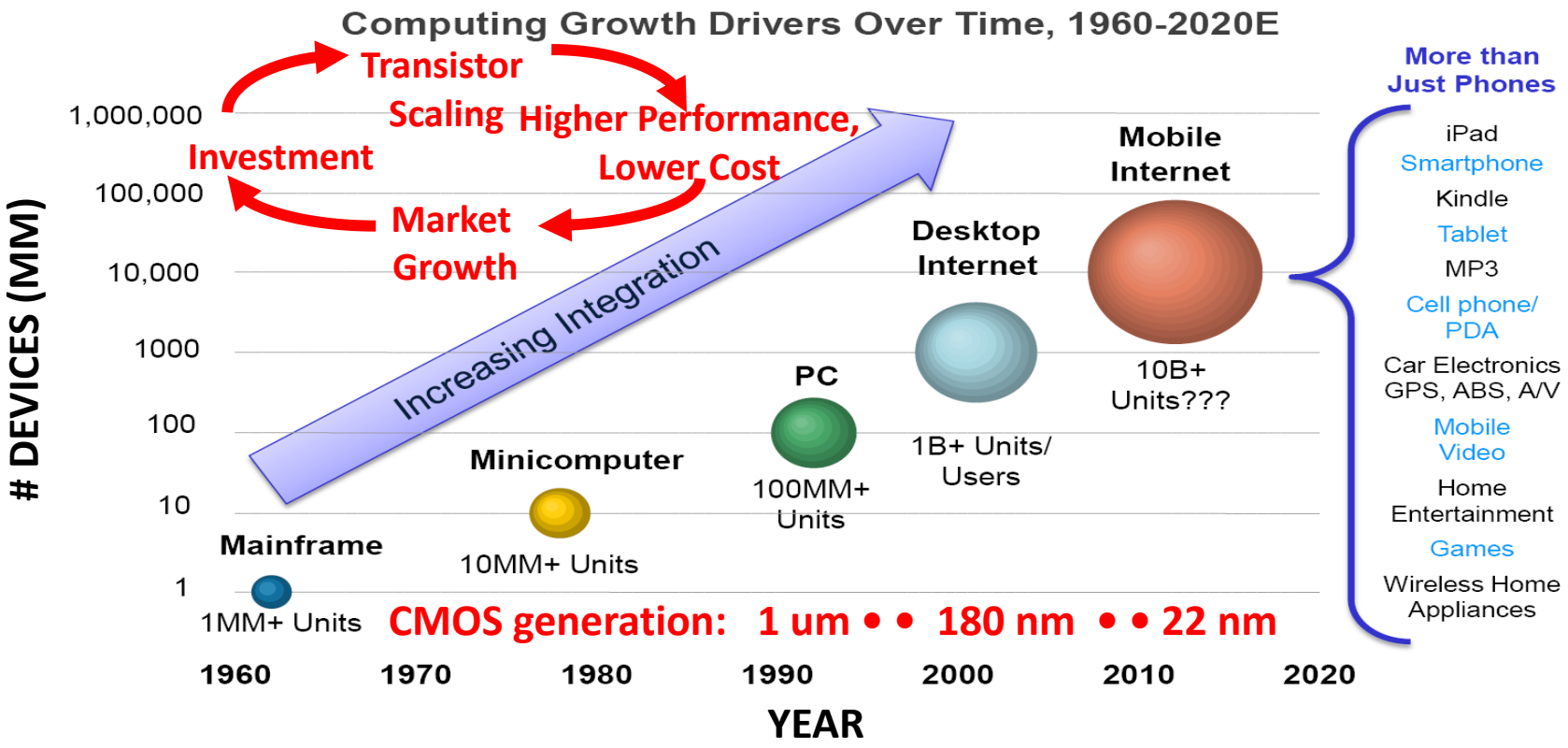


300mm Si wafer

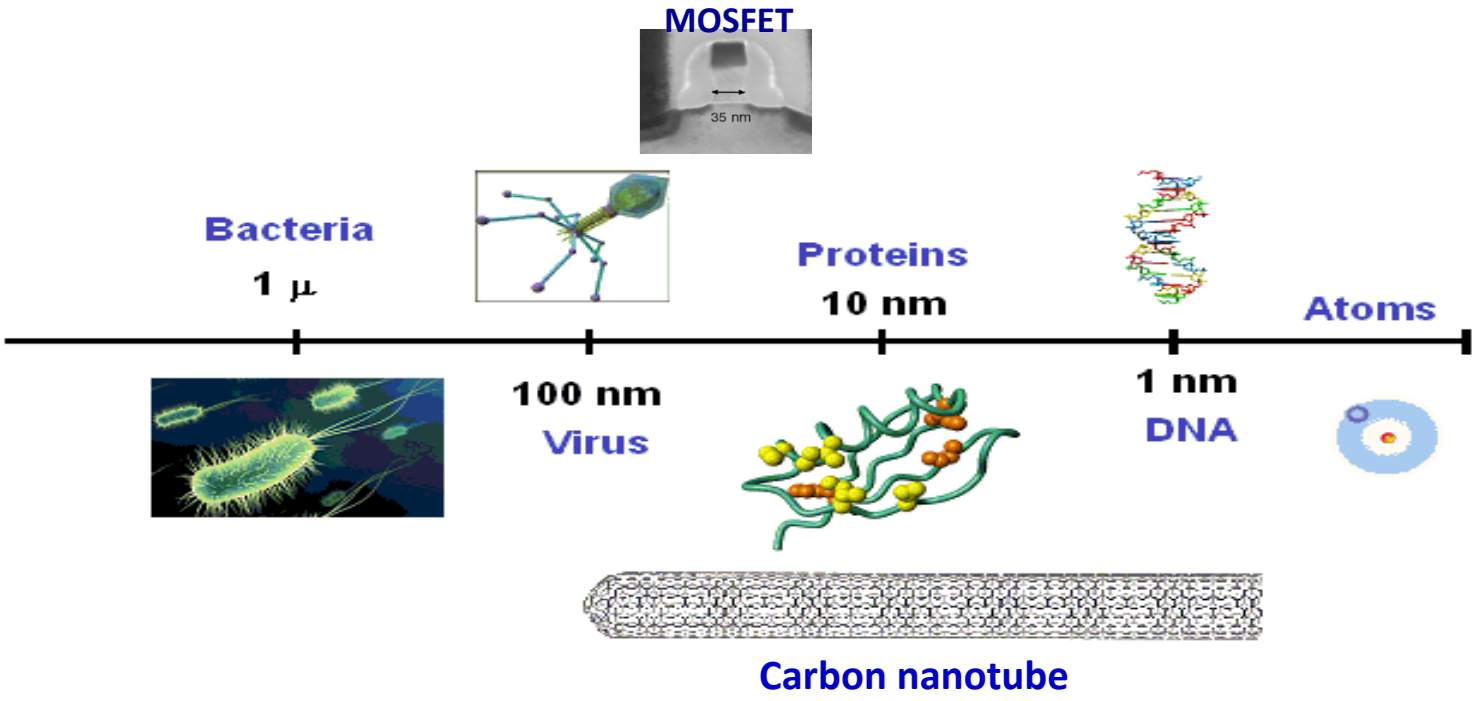
Moore's Law: The # of devices on a chip doubles every ~2 yrs, for the same chip price.

Impact of Moore's Law

http://www.morganstanley.com/institutional/techresearch/pdfs/2SETUP_12142009_RI.pdf



The Nanometer Size Scale



1 micrometer (1 μm) = 10^{-4} cm; 1 nanometer (nm) = 10^{-7} cm

Lecture 1

OUTLINE

- Important Quantities
- Semiconductor Fundamentals
  - General material properties
  - Crystal structure
  - Crystallographic notation
  - Electrons and holes

Reading: Pierret 1.1-1.2, 2.1; Hu 1.1-1.2

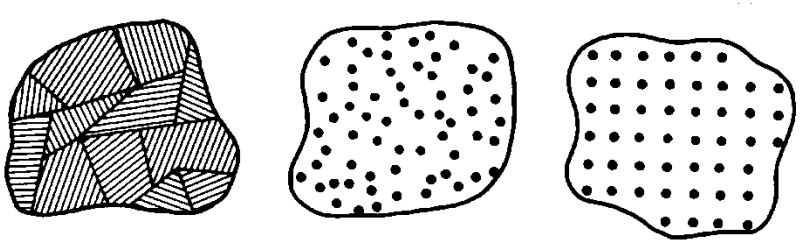
Important Quantities

- Electronic charge, $q = 1.6\times10^{-19}$ C
- Permittivity of free space, $\varepsilon_o = 8.854\times10^{-14}$ F/cm
- Boltzmann constant, $k = 8.62\times10^{-5}$ eV/K
- Planck constant, $h = 4.14\times10^{-15}$ eV·s
- Free electron mass, $m_o = 9.1\times10^{-31}$ kg
- Thermal voltage kT/q = 26 mV at room temperature
- kT = 0.026 eV = 26 meV at room temperature
- $kTln(10)$ = 60 meV at room temperature

1 eV = 1.6 x 10^{-19} Joules

What is a Semiconductor?

- Low resistivity => “conductor”
- High resistivity => “insulator”
- Intermediate resistivity => “semiconductor”
  - conductivity lies between that of conductors and insulators
  - generally crystalline in structure for IC devices
    - In recent years, however, non-crystalline semiconductors have become commercially very important



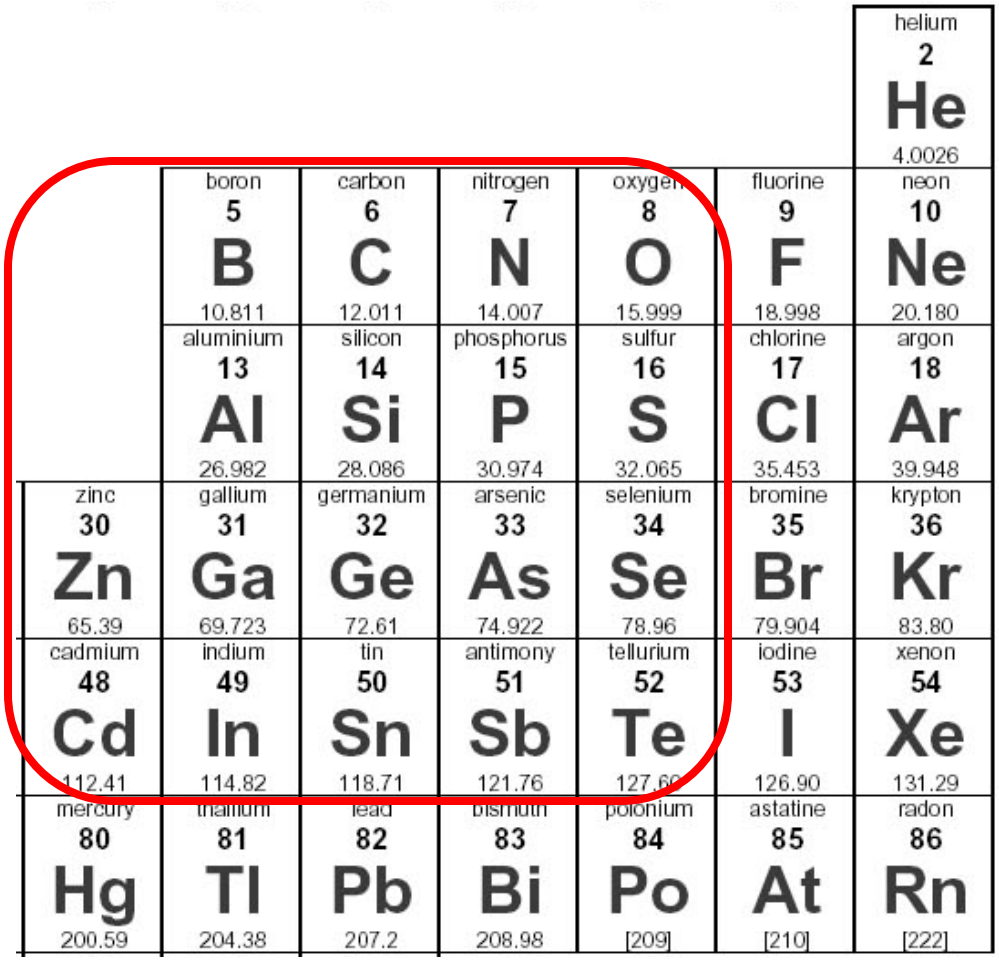
polycrystalline amorphous crystalline

Semiconductor Materials

Elemental:

Compound:

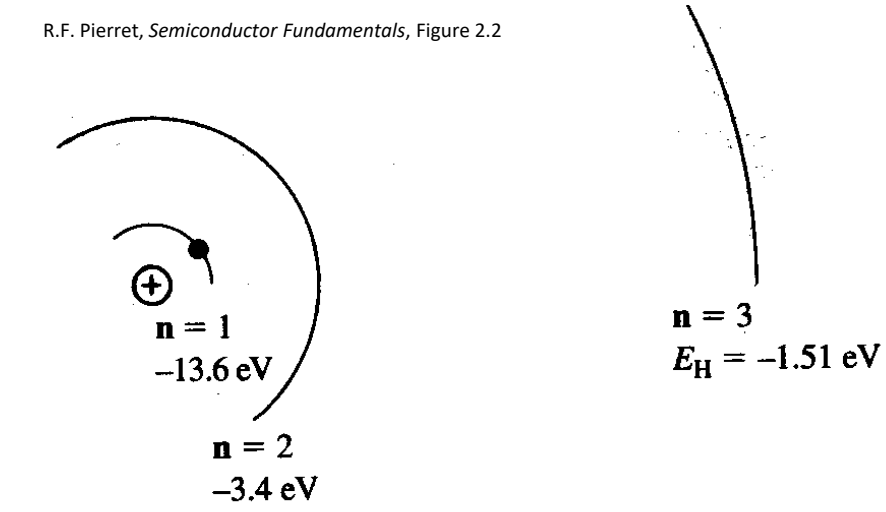
Alloy:

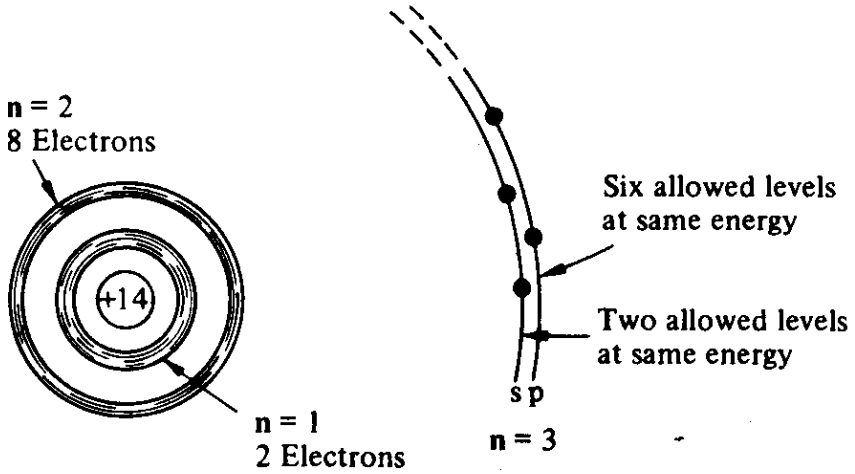


						helium 2 He 4.0026
	boron 5 B 10.811	carbon 6 C 12.011	nitrogen 7 N 14.007	oxygen 8 O 15.999	fluorine 9 F 18.998	neon 10 Ne 20.180
	aluminium 13 Al 26.982	silicon 14 Si 28.086	phosphorus 15 P 30.974	sulfur 16 S 32.065	chlorine 17 Cl 35.453	argon 18 Ar 39.948
zinc 30 Zn 65.39	gallium 31 Ga 69.723	germanium 32 Ge 72.61	arsenic 33 As 74.922	selenium 34 Se 78.96	bromine 35 Br 79.904	krypton 36 Kr 83.80
cadmium 48 Cd 112.41	indium 49 In 114.82	tin 50 Sn 118.71	antimony 51 Sb 121.76	tellurium 52 Te 127.60	iodine 53 I 126.90	xenon 54 Xe 131.29
mercury 80 Hg 200.59	thallium 81 Tl 204.38	lead 82 Pb 207.2	bismuth 83 Bi 208.98	polonium 84 Po [209]	astatine 85 At [210]	radon 86 Rn [222]

From Hydrogen to Silicon

R.F. Pierret, *Semiconductor Fundamentals*, Figure 2.2





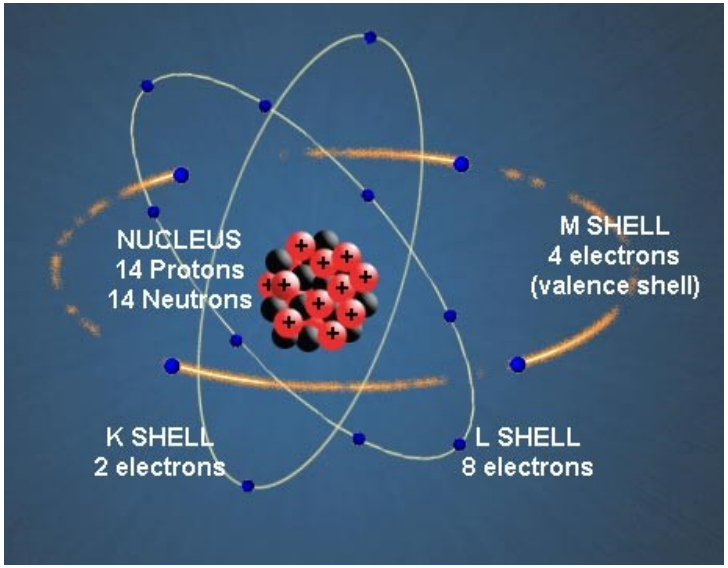
Z	Name	# of Electrons						Notation
		1	2		3			
		1s	2s	2p	3s	3p	3d	
1	H	1						$1s^1$
2	He	2						$1s^2$
3	Li	2	1					$1s^2\ 2s^1$
4	Be	2	2					$1s^2\ 2s^2$
5	B	2	2	1				$1s^2\ 2s^2\ 2p^1$
6	C	2	2	2				$1s^2\ 2s^2\ 2p^2$
7	N	2	2	3				$1s^2\ 2s^2\ 2p^3$
8	O	2	2	4				$1s^2\ 2s^2\ 2p^4$
9	F	2	2	5				$1s^2\ 2s^2\ 2p^5$
10	Ne	2	2	6				$1s^2\ 2s^2\ 2p^6$
11	Na	2	2	6	1			$1s^2\ 2s^2\ 2p^6\ 3s^1$
12	Mg	2	2	6	2			$1s^2\ 2s^2\ 2p^6\ 3s^2$
13	Al	2	2	6	2	1		$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$
14	Si	2	2	6	2	2		$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$
15	P	2	2	6	2	3		$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$
16	S	2	2	6	2	4		$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$
17	Cl	2	2	6	2	5		$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$
18	Ar	2	2	6	2	6		$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

The Silicon Atom

- 14 electrons occupying the first 3 energy levels:
  - 1s, 2s, 2p orbitals filled by 10 electrons
  - 3s, 3p orbitals filled by 4 electrons

To minimize the overall energy, the 3s and 3p orbitals hybridize to form 4 tetrahedral 3sp orbitals

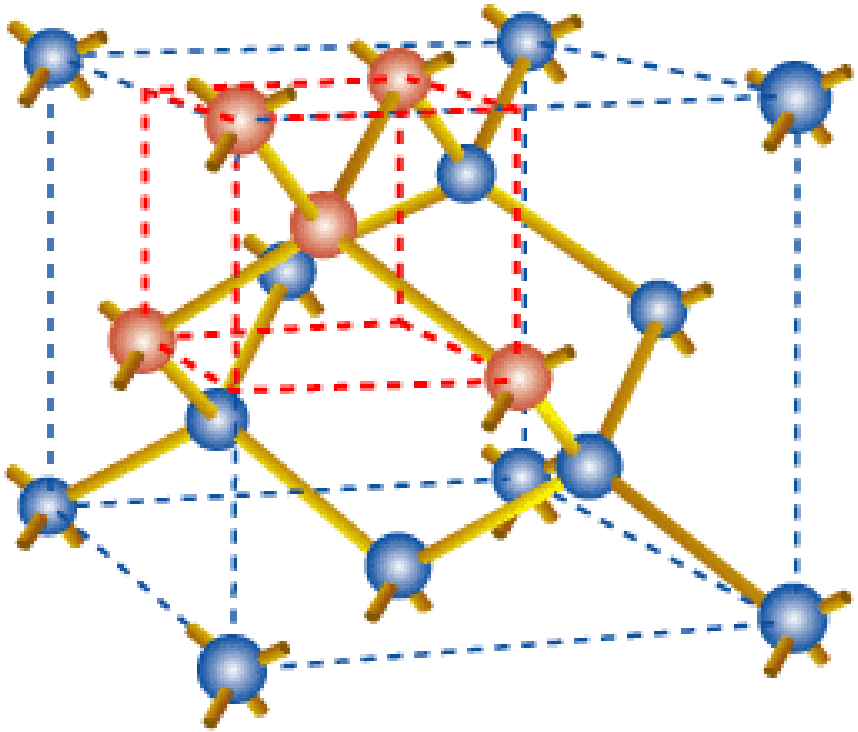
Each has one electron and is capable of forming a bond with a neighboring atom



http://www.learnabout-electronics.org/semiconductors_01.php

The Si Crystal

http://www.daviddarling.info/encyclopedia/S/AE_silicon.html



- **Each Si atom has 4 nearest neighbors**
  - **“diamond cubic” lattice**
  - **lattice constant = 5.431Å**

How Many Silicon Atoms per cm^3?

- Total number of atoms within a unit cell:

  Number of atoms completely inside cell:

  Number of corner atoms (1/8 inside cell):

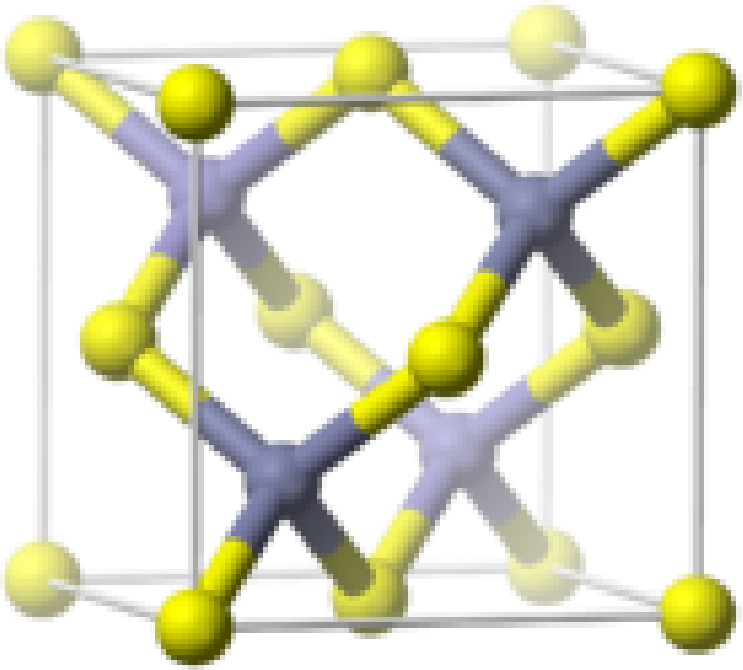
  Number of atoms on the faces (1/2 inside cell):

- Cell volume: $(0.543\ nm)^3$

- **Density of silicon atoms:**

Compound Semiconductors

http://en.wikipedia.org/wiki/Aluminium_gallium_arsenide

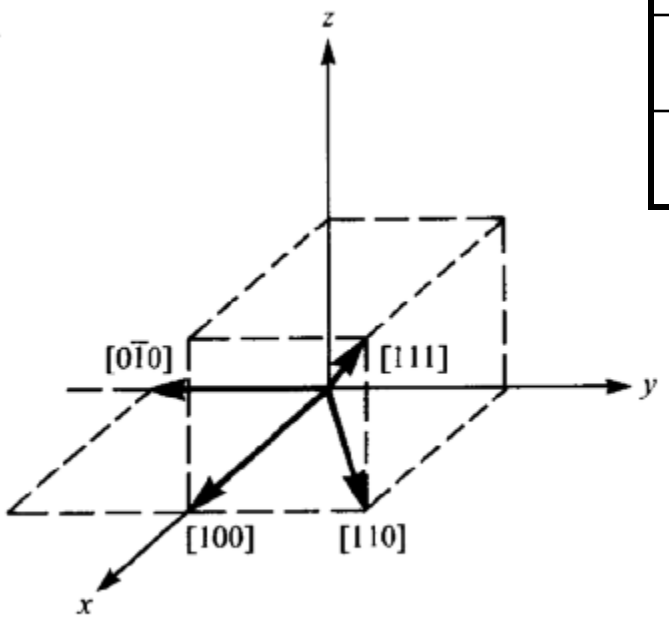


- "zincblende" structure
- III-V compound semiconductors: GaAs, GaP, GaN, *etc.*
  - ✓ important for optoelectronics and high-speed ICs

Crystallographic Notation

Miller Indices:

Notation	Interpretation
(*h k l*)	crystal plane
{ *h k l* }	equivalent planes
[*h k l*]	crystal direction
< *h k l* >	equivalent directions



Sample direction vectors and their corresponding Miller indices.

***h*: inverse *x*-intercept of plane**

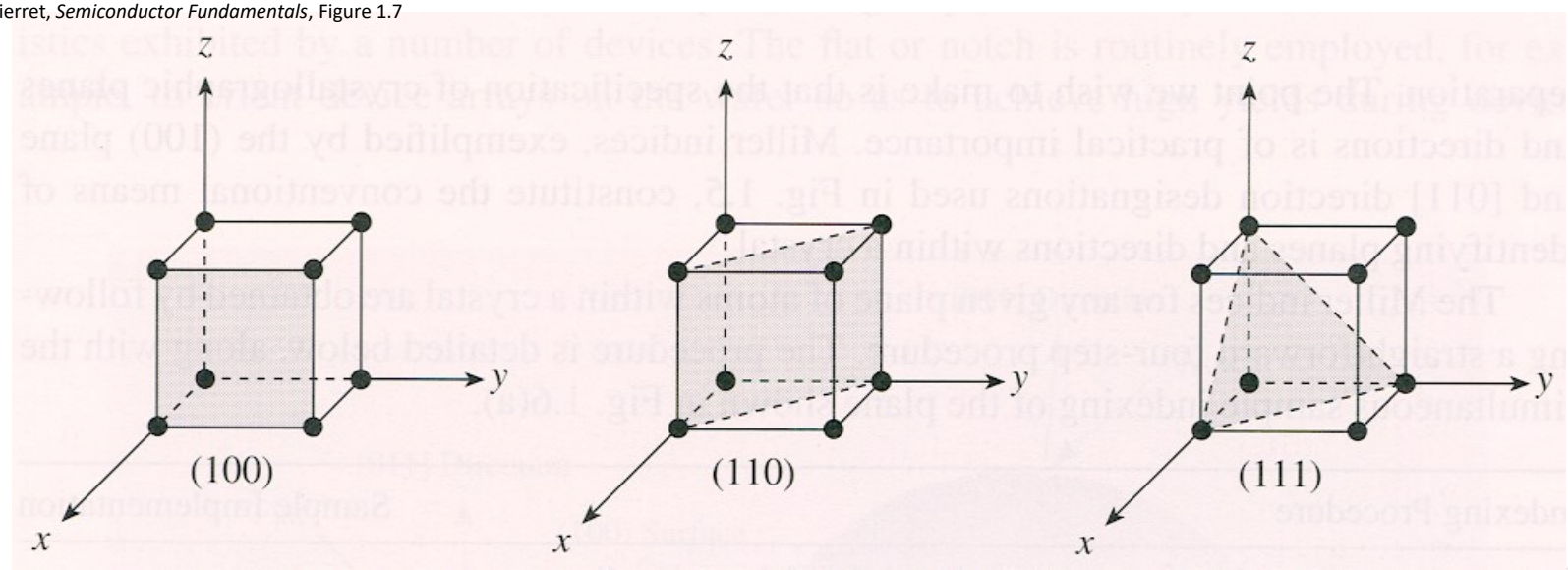
***k*: inverse *y*-intercept of plane**

***l*: inverse *z*-intercept of plane**

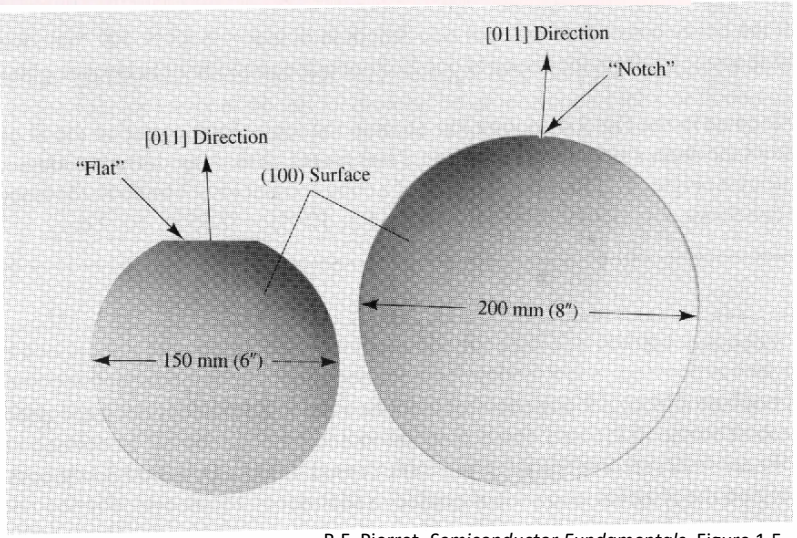
(Intercept values are in multiples of the lattice constant; *h*, *k* and *l* are reduced to 3 integers having the same ratio.)

Crystallographic Planes and Si Wafers

R.F. Pierret, *Semiconductor Fundamentals*, Figure 1.7



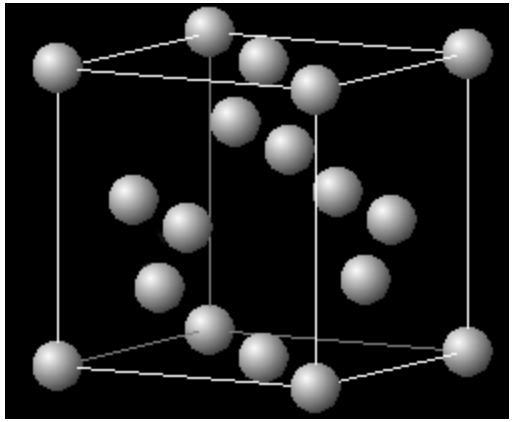
Silicon wafers are usually cut along a {100} plane with a flat or notch to orient the wafer during IC fabrication:



R.F. Pierret, *Semiconductor Fundamentals*, Figure 1.5

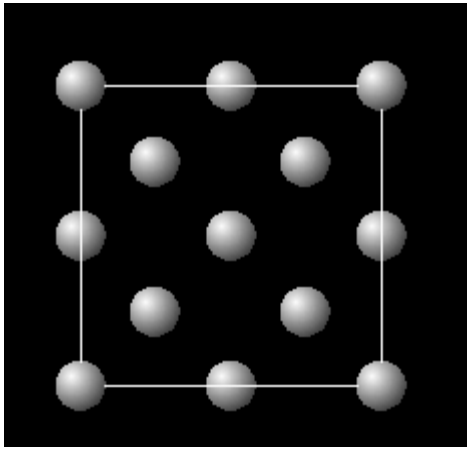
Crystallographic Planes in Si

Unit cell:

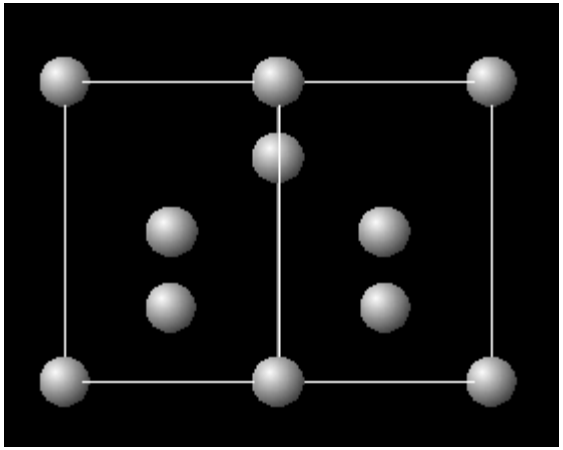


http://jas.eng.buffalo.edu/education/solid/unitCell/home.html

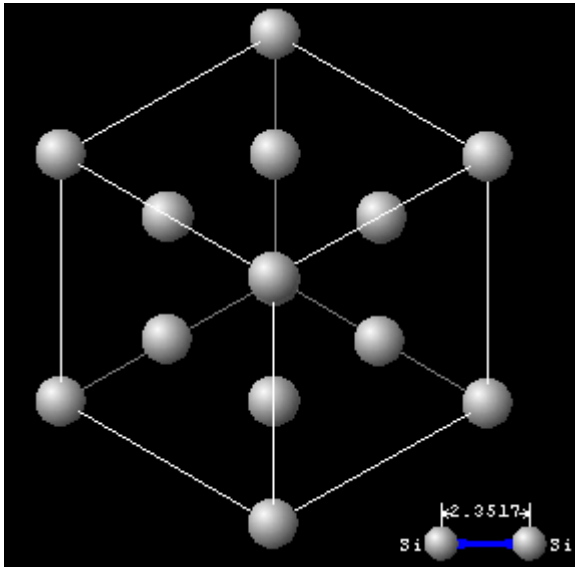
View in <100> direction



View in <110> direction



View in <111> direction



Electronic Properties of Si

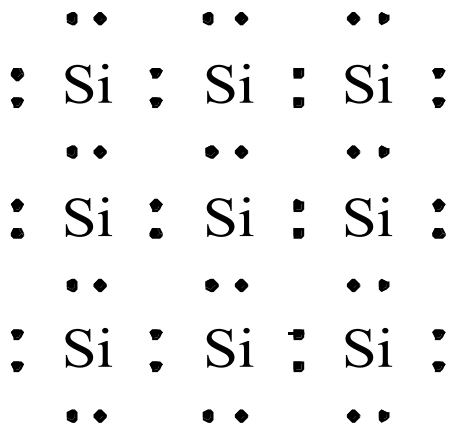
- **Silicon is a semiconductor material.**
  - Pure Si has relatively high electrical resistivity at room temp.

- **There are 2 types of mobile charge-carriers in Si:**
  - ***Conduction electrons*** are negatively charged
  - ***Holes*** are positively charged

- **The concentration (#/cm^3) of conduction electrons & holes in a semiconductor can be changed:**
  1. **by changing the temperature**
  2. **by adding special impurity atoms (*dopants*)**
  3. **by applying an electric field**
  4. **by irradiation**

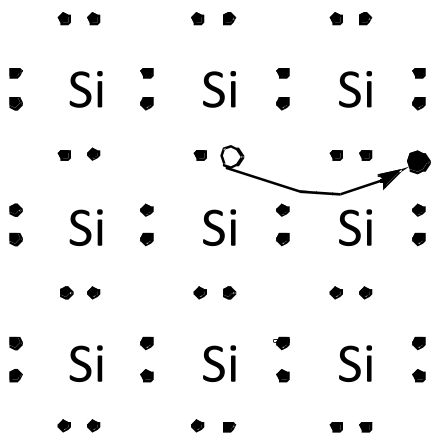
Electrons and Holes (Bond Model)

2-D representation of Si lattice:



C. C. Hu, *Modern Semiconductor Devices for ICs*, Figure 1-4

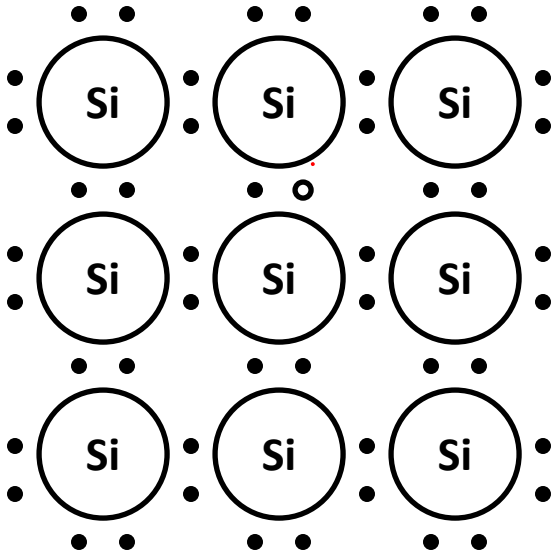
When an electron breaks loose and becomes a ***conduction electron***, a ***hole*** is also created.



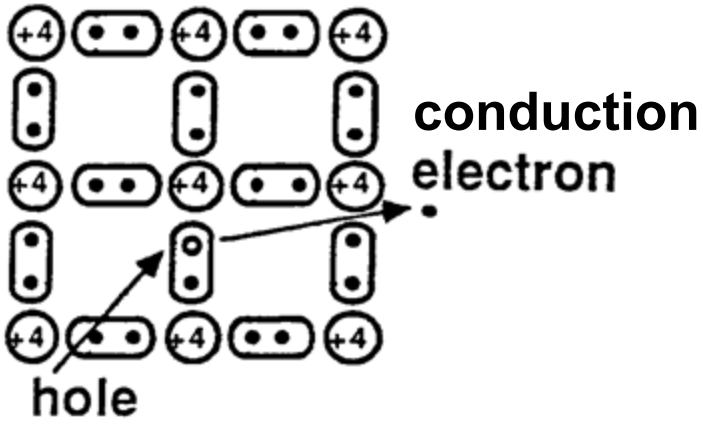
C. C. Hu, *Modern Semiconductor Devices for ICs*, Figure 1-5a

The Hole as a Positive Mobile Charge

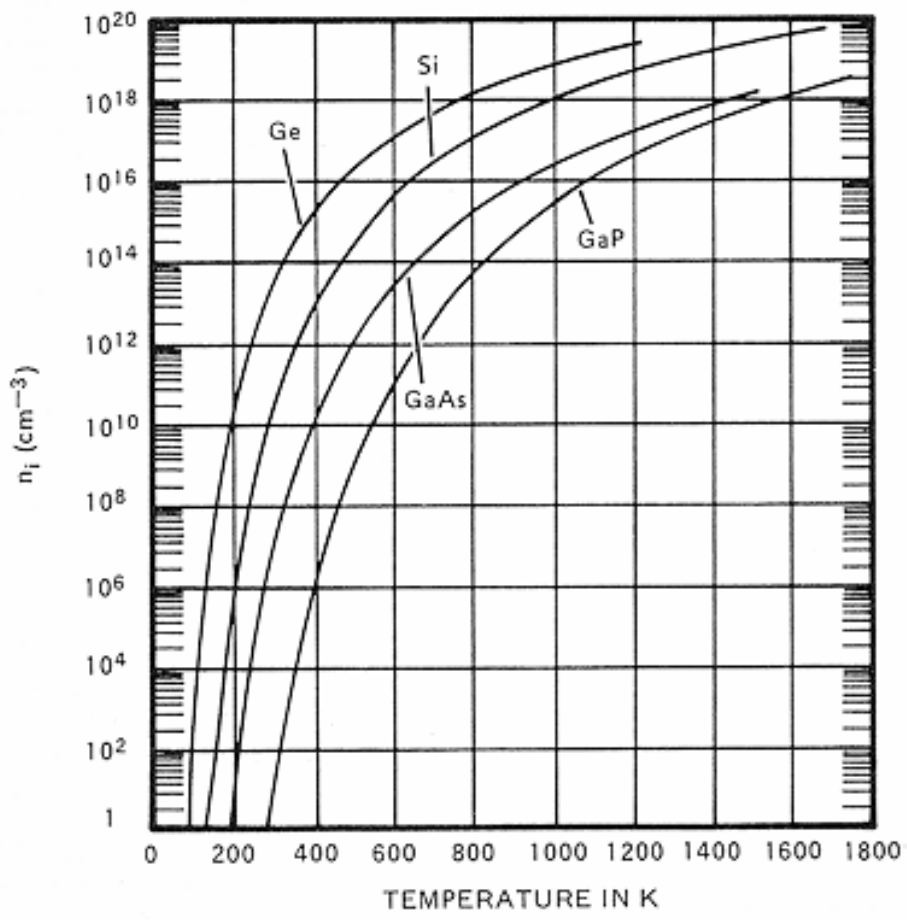
- **Positive charge is associated with a half-filled covalent bond**
  - Moves when an electron from a neighboring covalent bond fills it



Intrinsic Carrier Concentration, n_i



- At temperatures > 0 K, some electrons will be freed from covalent bonds, resulting in electron-hole pairs.



For Si: $n_i \cong 10^{10}$ cm^{-3} at room temperature

Definition of Terms

$n \equiv$ number of electrons/cm^3

$p \equiv$ number of holes/cm^3

$n_i \equiv$ intrinsic carrier concentration

In a pure semiconductor,

$$n = p = n_i$$

Summary

- Crystalline Si:
  - 4 valence electrons per atom
  - diamond lattice (each atom has 4 nearest neighbors)
  - **atomic density = 5 x 10^{22} atoms/cm^3**
  - **intrinsic carrier concentration n_i = 10^{10} cm^{-3}**
  - Miller indices are used to designate planes and directions within a crystalline lattice
- In a pure Si crystal, conduction electrons and holes are formed in pairs.
  - Holes can be considered as positively charged mobile particles.
  - Both holes and electrons can conduct current.

Acknowledgement

- Professor Tsu-Jae King Liu (UC Berkeley) EE130/230A notes
- Modern Semiconductor Devices for Integrated Circuits by C. Hu (Prentice Hall, 2009)
- Semiconductor Device Fundamentals by R. F. Pierret (Addison Wesley, 1996)